



Phabel A. Lopez-Delgado

Computational Biologist & Data Scientist

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Online Profiles

- GitHub
- Google Scholar
- LinkedIn
- ORCID

Languages

- Spanish (Native) ● ● ● ● ●
- English (Fluent – C1) ● ● ● ● ●
- German (Basic – A2) ● ● ● ● ●
- Italian (Basic – A2) ● ● ● ● ●

Research Experience

Biological Computational Research

2025 – present **TMO: Temporal Multi-Omics** Latest Project
Developed a deep learning framework using asymmetric cross-modal attention to learn signed, cell-state-conditional regulatory lags ($\Delta\tau$) from paired single-cell ATAC+RNA data. Achieved Lag Concordance Scores (LCS) 0.988–0.999, outperforming symmetric baseline (< 0.11). Validated on four 10x Multiome datasets with ChIP-seq and Perturb-seq. Open-source Python package `tmopy`. DOI: 10.64898/2026.06.08.730880

Clinical Research Collaborations

2024 – present **Research Consultant & Data Analyst** ISSSTE Hospital
Co-designed and analysed 4 pilot RCTs in postoperative pain and septic shock. Responsible for statistical analysis (LMM, ROC, logistic regression), data visualisation, and manuscript preparation. Co-author of 4 preprints.

Research Internships

- 2024 **Computational Epigenomics Lab** Helmholtz Munich / LMU
Multi-omic single-cell dynamics. Supervised by Prof. Dr. Maria Colomé-Tatché and M.Sc. Gabriele Malagoli.
- 2023 **Epigenetics, Stem Cells & AI** Helmholtz Munich
Analysed in vitro models of embryonic development with ML and physics. Supervised by Dr. Antonio Scialdone and Dr. Gabriele Lubatti.
Competitive selection: Only 5 students selected internationally; I was the only undergraduate participant and the only one from outside Europe.
- 2022–2023 **Transcription, Epigenetics & Development Lab** UNAM – IBT
Supervised by Dr. Viviana del Carmen Valadez Graham (Prof. Dr. Mario Zurita).
- 2021–2023 **Systems Biology Lab** UNAM – CCG
Five semesters in Systems Biology & Synthetic Biology (Dr. Julio Augusto Freyre González).

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Technical Skills

-  Python, R, C, C++, Bash, SQL, LaTeX, HTML, Excel
-  PyTorch, TensorFlow, Keras, scikit-learn, XGBoost
-  scanpy, anndata, Seurat, epiScanpy, ChIP-seq, Perturb-seq
-  ggplot2, ggstatsplot, matplotlib, seaborn, plotly
-  Git, GitHub, Linux/Bash, reproducible research
-  Transformers, attention, GNNs, RNNs, ANNs
-  Supervised & Unsupervised ML, clustering, PCA, t-SNE, UMAP
-  Claude, DeepSeek, Gemini, ChatGPT, NotebookLLM

About Me

Computational biologist and data scientist with a Licenciatura in Genomic Sciences (UNAM). Author of 5 preprints (4 clinical, 1 computational) and developer of TMO, a transformer-based framework for regulatory lag inference. Experienced in interdisciplinary research, statistical consulting, and collaboration with clinicians. Also passionate about sports science and dance.

Education

Undergraduate

2021 – 2025	Licenciatura en Ciencias Genómicas (Undergraduate Program on Genomic Sciences) Cédula Profesional (Professional License): 15699774 Thesis: “Multi-Omic Single-Cell Trajectory Inference with Multi-Modal Diffusion Maps in Cellular Dynamics” <i>Global Grade: 9.75/10</i>	UNAM
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Graduate Diplomas & Courses

2025–2026	Advanced Data Analysis (150h) Project: Python ShinyApp for Single-Cell Analysis	UNAM
2025	Applied Artificial Intelligence (195h) Grade 9.9/10. Project: Prediction of Cell Types from Single-Cell Data with AI	UNAM
2025	Pharmacogenomics	UNAM / INMEGEN
2026	RENASCEED for Proficient Sports Trainers Ongoing – 2/5 modules completed (first generation of the program in Mexico)	CONADE – ENED

Preprints and Publications

<i>Submitted</i>	TMO: Asymmetric cross-modal attention for learning cell-state-dependent regulatory lags from single-cell multiomic data <i>Lopez-Delgado PA, Delgado-Carlo MM</i> bioRxiv. 2026. doi:10.64898/2026.06.08.730880
<i>Submitted</i>	Prognostic value of mean platelet volume in septic shock: a retrospective cohort study <i>Trujillo-Vega F, Lopez-Delgado PA</i> medRxiv. 2026. doi:10.64898/2026.05.29.26354453
<i>Submitted</i>	Sphenopalatine ganglion block for post-dural puncture headache: a pilot randomized controlled trial <i>Everardo-Salazar G, Lopez-Delgado PA, Delgado-Carlo MM</i> medRxiv. 2026. doi:10.64898/2026.05.06.26352338
<i>Submitted</i>	Analgesic equivalence of NSAIDs and a weak opioid in acute postoperative pain following minimally invasive surgery: a pilot randomized controlled trial <i>Vallejo-Mora PE, Lopez-Delgado PA, Delgado-Carlo MM</i> medRxiv. 2026. doi:10.64898/2026.05.03.26352343
<i>Submitted</i>	Oral premedication with tapentadol versus pregabalin for acute postoperative pain in lower limb surgery under neuraxial anesthesia: a pilot randomized controlled trial <i>Escalona-Arroyo MDR, Lopez-Delgado PA, Delgado-Carlo MM</i> medRxiv. 2026. doi:10.64898/2026.05.14.26353251

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Hard Skills (Detailed)

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Python

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R

Single-cell multi-omics

Deep Learning (PyTorch)

Data Visualisation

Git/GitHub

Transformers / Attention

Supervised ML

Unsupervised ML

Professional Experience

2025 – present	Research Consultant & Data Analyst ISSSTE Hospital Provided statistical consulting for ~15 clinical research projects in anesthesiology and traumatology. Designed analysis plans, performed data wrangling, visualisation, and modelling (LMM, ROC, logistic regression). Co-authored 4 preprints. First author of 1 preprint.
2022 – 2024	Teaching Assistant UNAM <i>Functional Genomics II – Epigenetics</i> (2024-2): Assisted Prof. Dr. Félix Recillas Targa. <i>Scientific Computing</i> (2023-2): Assisted Dr. Julio Augusto Freyre González. <i>Programming Principles</i> (2022-1): Assisted Dr. Julio Augusto Freyre González.

Awards and Recognitions

2023	Competitive Internship Selection Helmholtz Munich Only 5 students selected internationally; I was the only undergraduate participant and the only one from outside Europe.
2025	Grade 9.9/10 UNAM – Applied AI Diploma
2021–2025	Global Grade 9.75/10 UNAM – Genomic Sciences

Conferences, Workshops, and Seminars (Selected)

2026	Congress in Autistic Spectrum Disorder	UNAM-INMEGEN
2026	Symposium of Genomic Sciences and Cellular Physiology	IFC-UNAM
2025	1st International Congress on AI: Challenges and Opportunities	UNAM
2025	International Congress in Sports Science, Physiotherapy, and Health	Universidad CDEFIS
2025	Workshops: AI for Health Sciences, ML for Movement Analysis, Graph Deep Learning	UNAM / IPN
2024	6th Munich International Chromatin Symposium	LMU
2024	Munich Epigenetics Spotlight Meeting; Epigenetics Retreat	Helmholtz Munich

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Soft Skills

 Interdisciplinary Collaboration

 Scientific Communication

 Independence & Initiative

 Problem-Solving

 Adaptability

 Teaching & Mentoring

 Discipline & Perseverance

Interests

-  Sports Science & Physical Training
-  Dance: Bachata (Advanced), Zouk (Intermediate)
-  Music, Travel, Teaching

Contact Me

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Google Scholar: scholar.google.com/...

Extra-Curricular Activities

Sports Science and Physical Training

2026	RENASCEED Diploma for Proficient Sports Trainers (2/5 modules)	CONADE
2025	Seminars: Core Biomechanics, Advanced Resistance Training, Hypertrophy Planning	Various
2025	Congresses: Sports Science, Nutrigenomics, Sports Supplementation	Various

Dance and Creative Arts

2023–present	Bachata – Advanced	Grupo Esencia México
2024	Bavarian Bachata Congress; Munich Bachazouk Festival	International
2025–present	Brazilian Zouk – Intermediate	Zouk Vibes México

Data Analyst Portfolio (Selected Projects)

2026	Web App for Autistic Spectrum Disorder diagnosis	Client
2025	Pregabalin 150 mg vs. 75 mg for postoperative pain in radical mastectomy	Client
2025	Ultrasound-guided fascia iliaca block vs. local infiltration for hip surgery	Client
2025	Ondansetron vs. water co-loading for prevention of hypotension in cesarean section	Client
2024	Lymphocyte-neutrophil index vs. platelet-lymphocyte index as acute cholangitis predictors	Client
2023	Apneic oxygenation vs. denitrogenation in rapid sequence intubation	Client
2023	Intravenous haloperidol vs. ondansetron for nausea/vomiting prophylaxis in laparoscopic cholecystectomy	Client

Full Consulting Portfolio

2025	Intravenous butylhyoscine for prevention of bradycardia in geriatric patients	Client
2025	Nebulised lidocaine vs. intravenous infusion for control of hemodynamic changes during laryngoscopy	Client
2025	Bupivacaine + dexmedetomidine vs. bupivacaine alone in total abdominal hysterectomy	Client
2023	Ropivacaine 2% for postoperative analgesia	Client
2023	Nasal vs. intravenous dexmedetomidine for sedation in eye surgery	Client

References

Available upon request